Characterization Techniques for Herbal Products

Rakhee*, Jigni Mishra†, Raj K. Sharma*, Kshipra Misra‡

*Department of Chemistry, University of Delhi, Delhi, India, †Chemistry Division, Department of Biochemical Sciences (DBCS), Defence Institute of Physiology and Allied Sciences (DIPAS), Delhi, India, ‡Department of Biochemical Sciences (DBCS), Defence Institute of Physiology and Allied Sciences (DIPAS), Delhi, India

Abbreviations

^{13}C NMR	Carbon-13 nuclear magnetic resonance
^{1}H NMR	proton nuclear magnetic resonance
^{1}H NMR	proton nuclear magnetic resonance
2D NMR	two-dimensional nuclear magnetic resonance spectroscopy
ABTS^{+}	2,2′-azino-bis (3-ethylbenzothiazoline-6-sulphonic acid)
API	atmospheric pressure ionization
ASE	accelerated solvent extraction
BMOEA bst	bis (2 methoxyethyl) ammonium bis (tri-uoromethylsulfonyl) imide
CE-MS	capillary electrophoresis-mass spectrometry
***d*^{6}-DMSO**	hexadeuterodimethyl sulfoxide
DEPT-20	distortionless enhancement by polarization transfer-20 type ^{13}C NMR
DMAE	dynamic microwave-assisted extraction
DMEA oct	*N*,*N*-dimethylethanolammonium octanoate
DNA	deoxyribonucleic acid
DOSY	diffusion ordered-NMR spectroscopy
DPPH	2,2-dipheyl-1-picryhydrazyl
EOF	electroosmotic flow
ESI-MS	electrospray ionization-mass spectrometry
ESI-MS/MS	electrospray ionization-mass spectrometry/mass spectrometry
FC	flash chromatography
FRAP	ferric-reducing antioxidant power
FT-IR	Fourier transform-infrared spectroscopy
GC	gas chromatography
GC-FTIR	gas chromatography-Fourier transform infrared spectroscopy
GC-MS	gas chromatography-mass spectrometry
H1-H1	proton-proton coupled nuclear magnetic resonance
HILIC	hydrophilic interaction liquid chromatography
HPLC	high pressure liquid chromatography
HPLC-^{1}H NMR	high-pressure liquid chromatography-proton-nuclear magnetic resonance

Management of High Altitude Pathophysiology. https://doi.org/10.1016/B978-0-12-813999-8.00009-4

HPLC-DAD/UV	high-performance liquid chromatography-diode array detector/ultraviolet spectroscopy
HPLC-DAD-ESI-MS/MS	high-performance liquid chromatography-diode array detector-electron spray ionization-mass spectroscopy/mass spectroscopy
HPLC-IR	high-pressure liquid chromatography-infrared spectroscopy
HPLC-NMR-MS	high-pressure liquid chromatography-nuclear magnetic resonance-ultraviolet spectroscopy
HPLC-UV	high-pressure liquid chromatography-ultraviolet spectroscopy
HPLC-UV-IR	high-pressure liquid chromatography-ultraviolet spectroscopy-infrared spectroscopy
HPTLC	high-performance thin layer chromatography
IR	infrared spectroscopy
JRES	J-resolved spectroscopy
KBr	potassium bromide
LAMP	loop-mediated isothermal amplification
LC-FTIR	liquid chromatography-Fourier transform infrared spectroscopy
LC-MS	liquid chromatography-mass spectroscopy
LC-MS-MS	liquid chromatography-mass spectroscopy-mass spectroscopy
LC-NMR-MS	liquid chromatography-nuclear magnetic resonance-mass spectroscopy
LC-PDA-MS	liquid chromatography-photo diode array-mass spectrometry
LC-PDA-NMR-MS	liquid chromatography-photo diode array-nuclear magnetic resonance-mass spectrometry
LC-TSP-MS	liquid chromatography-thermospray-mass spectrometry
LC-UV-MS	liquid chromatography-ultraviolet-mass spectrometry
LPLC	low-pressure liquid chromatography
LVI-GC-MS	large volume injection-gas chromatography-mass spectrometry
MAE	microwave-assisted extraction
MALDI-TOF/TOF	matrix-assisted laser desorption/ionization-time of flight mass spectrometer
MDA	malondialdehyde
MS	mass spectrometry
Multiplex PCR	multiplex polymerase chain reaction
NMR	nuclear magnetic resonance
NPMAE	nitrogen-protected microwave-assisted extraction
PCR	polymerase chain reaction
PPC	preparative planar chromatographic
ROD	silica rod
RP-HPLC	reversed-phase high-performance liquid chromatography
SARP	several ankyrin repeat protein
Semi-prep-HPLC	semi-preparative high-performance liquid chromatography
SFE	supercritical fluid extraction
SOD	superoxide dismutase
SPE-LC-MS	solid-phase extraction-liquid chromatography-tandem mass spectrometry
SSCP-PCR	single-strand conformation polymorphism-polymerase chain reaction
TLC	thin-layer chromatography
TMS	tetra methyl silane
TSP	thermospray soft ionization technique
UAE	ultrasound-assisted extraction
UMAE	ultrasonic microwave-assisted extraction
UV	ultraviolet spectroscopy

9.1 INTRODUCTION

Despite extensive development in the field of extraction and characterization of medicinally valued and pharmacologically active natural products from various sources, there is a need to consolidate all the available characterization data in one place. Several extraction methods using different solvents, ranging from low to high polarity or vice versa, have been used to isolate pure bioactive compounds (Adou, 2005). Separation and identification of the bioactive ingredients of the herbs have been carried out by both conventional techniques, including qualitative methods such as proximate and phytochemical analysis, and nonconventional techniques, including GC, HPLC, LC, NMR, UPLC, GC-MS, HPLC-MS, HPLC-SPE-NMR, HPLC-UV-DAD, HPTLC-MS, UP-HPLC, and UPLC-DAD-TOF-MS. (Bucar et al., 2013; Chauhan, 2014; Gođevac et al., 2015; Kang, 2012; Mahrous and Farag, 2015; Yan et al., 2007). This chapter majorly focuses on the details underlying these techniques, with special reference to their applications in characterizing herbs growing at high altitudes. Such herbal sources, including *Cordyceps sinensis*, *Ganoderma lucidum*, *Hippophae* sp., *Rhodiola rosea*, and *Valeriana wallichii*, are rich in bioactive metabolites that impart endurance to the extreme climatic conditions (high pressure, low oxygen, and low temperature) prevailing at HA (Bhardwaj et al., 2016; Geetha et al., 2005; Rakhee et al., 2016, 2017; Sharma et al., 2012). The most prominent metabolites in these herbal sources are flavonoids, lipids, phenolics, polysaccharides, proteins, saponins, sugars, and tannins. (Azmir et al., 2013). The structures of some of the most commonly occurring phytoconstituents are depicted in Fig. 9.1.

9.2 CHARACTERISTICS OF HA HIMALAYAN MEDICINAL PLANTS

The HA Himalayan regions, until recently, were less explored because of difficulties with accessibility and harsh climatic conditions. In the last few years, however, HA medicinal plants and herbs have gained substantial interest among researchers because of their unique and rich content of numerous bioactive constituents (Theis and Lerdau, 2003). Considering the abundance of medicinally important pharmacological compounds present in HA herbs, around 70%–80% of the world's population depend on drug preparations from such herbal sources to treat several health problems.

Some of these pharmacological constituents along with their bioactivities are summarized in Table 9.1.

Some of the medicinal plants and herbs, such as *Aconitum heterophyllum*, *Acorus calamus*, *Asparagus racemose*, *Atropa belladonna Linn.*, *Bergenia ligulate*, *Cinchona*

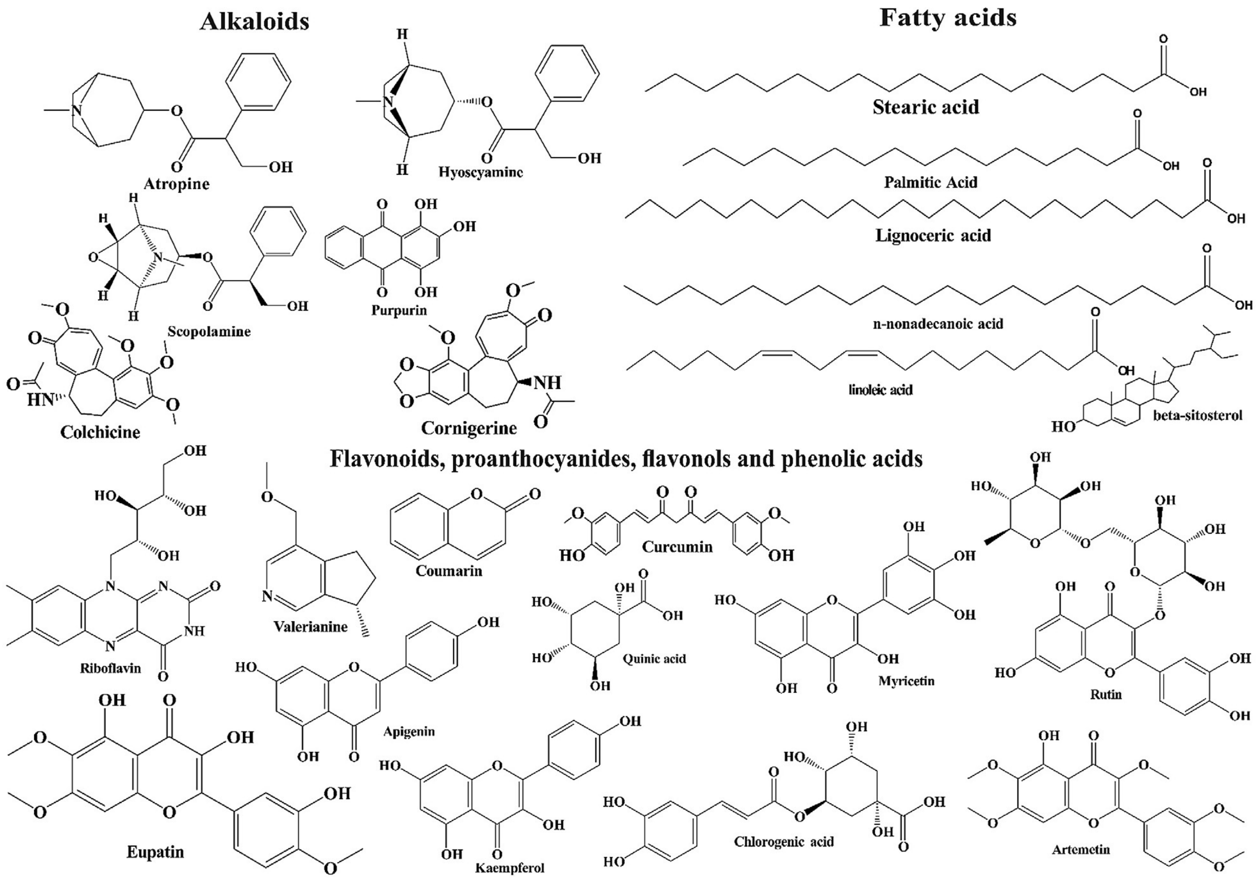


FIG. 9.1

Major classes of bioactive constituents found in medicinal plants.

(Continued)

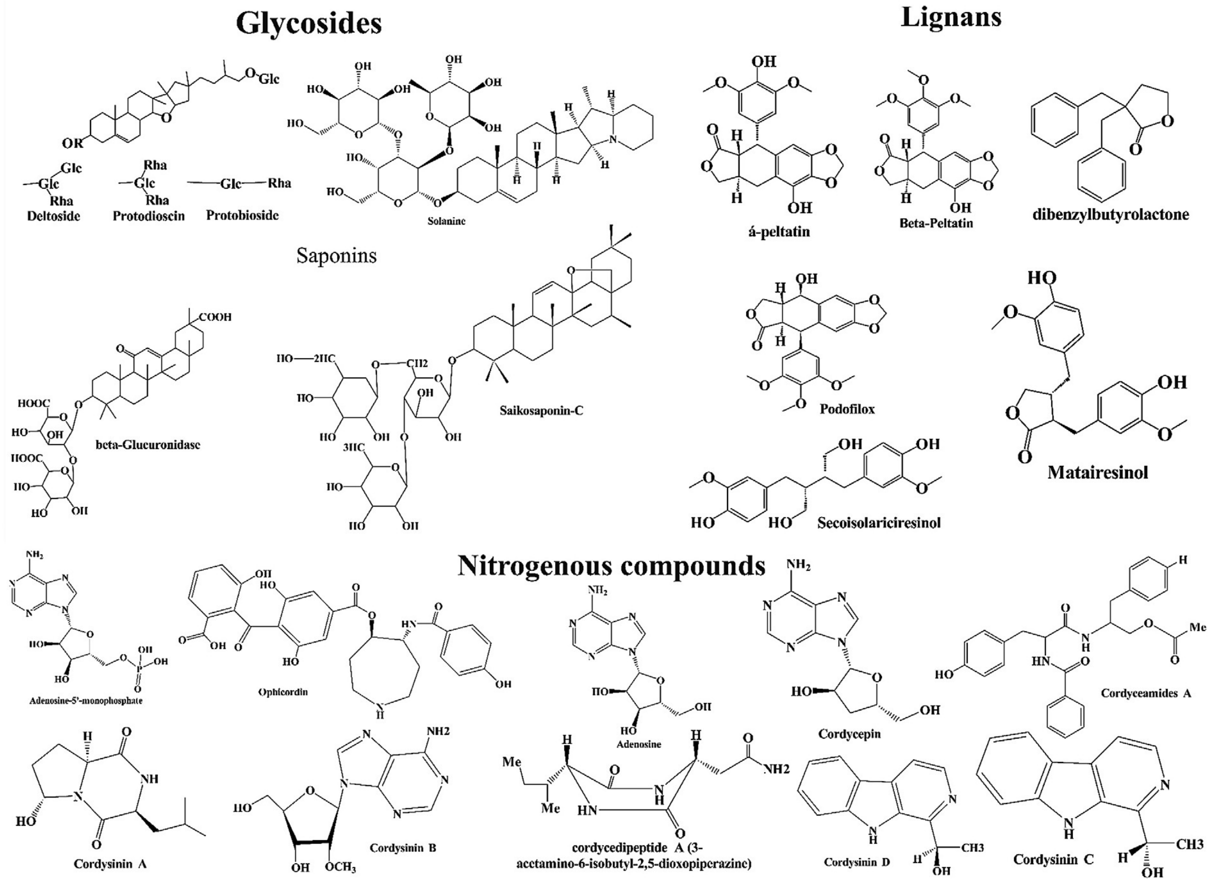


FIG. 9.1—cont'd

(Continued)

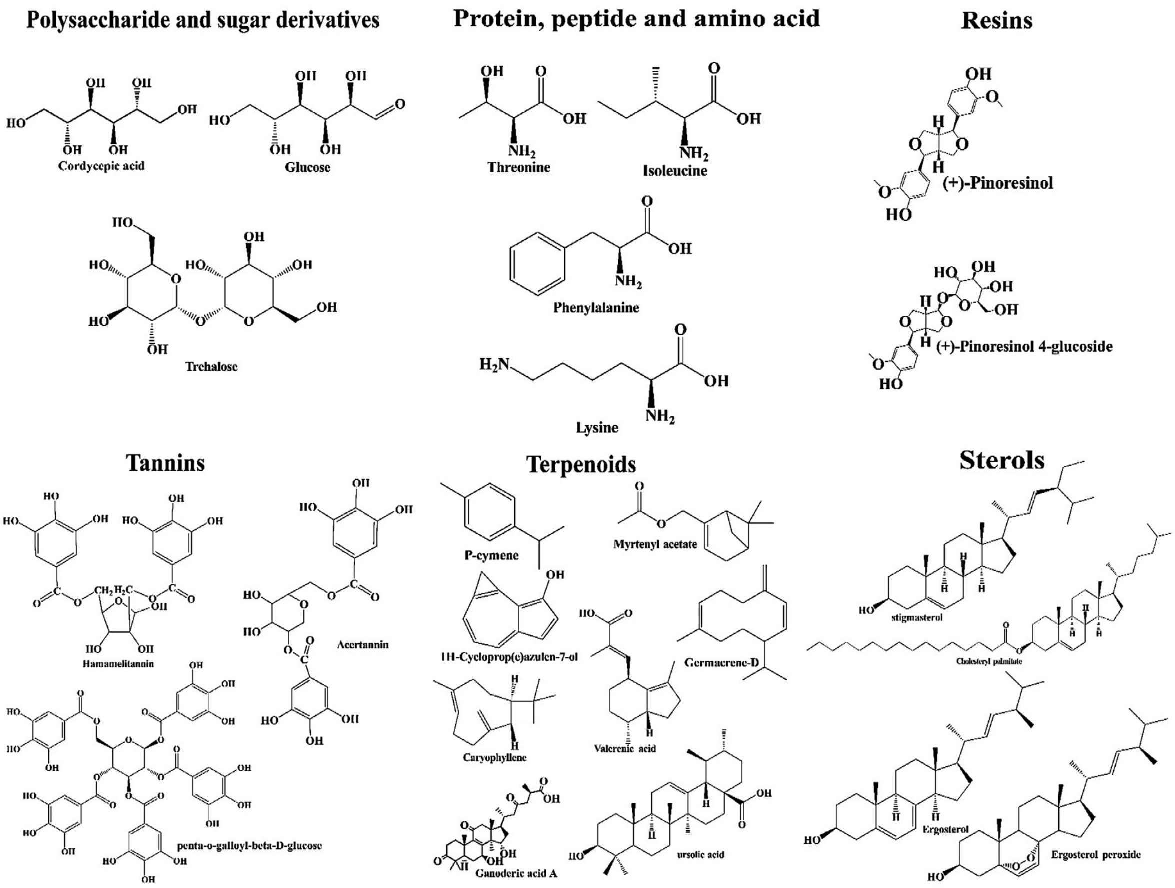

Polysaccharide and sugar derivatives
Cordycepic acid
Glucose
Trehalose
Protein, peptide and amino acid
Threonine
Isoleucine
Phenylalanine
Lysine
Resins
(+)-Pinoresinol
(+)-Pinoresinol 4-glucoside
Tannins
Hamamelitannin
Acertannin
penta-o-galloyl-beta-D-glucose
Terpenoids
P-cymene
Myrtenyl acetate
1H-Cycloprop(e)azulen-7-ol
Germacrene-D
Caryophyllene
Valerenic acid
Ganoderic acid A
ursolic acid
Sterols
stigmasterol
Ergosterol
Ergosterol peroxide

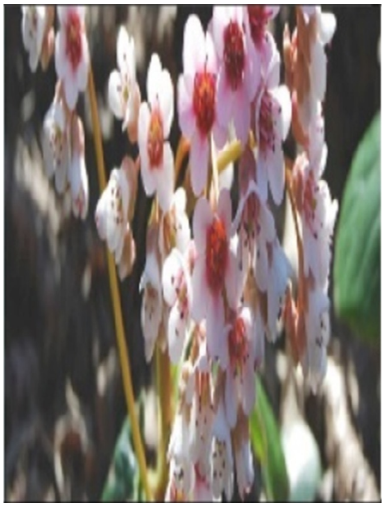
FIG. 9.1—cont'd

Table 9.1 High-Altitude Medicinal Plants With Their Active Components, Distribution, and Therapeutic Activities

Medicinal Plants	Images	Active Components	Distribution	Therapeutic Activities	References
Aconitum heterophyllum		Aconitine, benzoylmesaconine, mesaconitine, hypaconitine, hetidine, atidine	Uttaranchal, Himachal Pradesh, Sikkim (common in the subalpine and alpine zones of Himalayas at altitudes between 1800 and 4500)	Antiperiodic, anodyne, antidiabetic, antipyretic, narcotic, powerful sedative	Kaul (1997); http://senthuherbals.blogspot.in/2015/03/aconitum-heterophyllumadhividayam.html?view=snapshot
Acorus calamus		Myristic, palmitic, stearic, oleic, linoleic, arachidic, asarone, beta-asarone, (−)-4-terpineol, 2-allyl-5-ethoxy-4-methoxyphenol, epieudesmin, lysidine, (−)-spathulenol, borneol, furyl ethyl ketone, nonanoic acid, 2,2,5,5-tetramethyl-3-hexanol, bornyl acetate	Uttaranchal, Himachal Pradesh (found throughout the Indian Himalayas up to 6000 m and in Sri Lanka)	Intellect promoting, thermogenic, alexeteric	Balakumbahan et al. (2010)
Artemisia annua Linn.		Coumarin, apigenin, artemetin, clilorogenic acid, quinic acid, coumaric acid, myricetin, kaempferol, rutin, isorhanetin, eupatin, axillarin	Himachal Pradesh	Antimalarial, antibacterial, anticancer, antioxidant activity, antiasthma	Gupta et al. (2009), Chen et al. (2003), Kale et al. (2008); https://en.wikipedia.org/wiki/Artemisia_annua#cite_note-39

Continued

Table 9.1 High-Altitude Medicinal Plants With Their Active Components, Distribution, and Therapeutic Activities—cont'd

Medicinal Plants	Images	Active Components	Distribution	Therapeutic Activities	References
Asparagus racemosa		Steroidal saponins, shatavaroside A, shatavaroside B, filiasparoside C, shatavarins, immunoside, 8-methoxy-5,6,4′-trihydroxyisoflavone 7-O-β-D-glucopyranoside	Arunachal Pradesh, Meghalaya	Antiseptic, dysentry, diuretic	Hayes et al. (2008), Saxena and Chourasia (2001); https://en.wikipedia.org/wiki/Asparagus_racemosus
Atropa belladona Linn.		Atropine, l-hyoscyamine, l-hyoscine, 1-arginine, l-ornithine, scopolamine, hyoscyamine	Assam (District of Goalpara)	Antiinflammatory, anticholinergic/antispasmodic action, mild sedative	Mathews et al. (1990); Kaul (2010); Basumatary et al. (2004); https://en.wikipedia.org/wiki/Atropa_belladonna
Bergenia ligulata		Bergenin, tannic acid, gallic acid, stigmesterol, P-sitosterol, catechin, (+)-afzelechin, 1, 8-cineole, isovalaric acid, (+)-(6*S*)-parasorbic acid, arbutin, phytol, caryophyllene, damascenone, β-eudesmol, 3-methyl-2-buten-l-ol, (*Z*)-asarone, terpinen-4-ol, paashaanolactone	Uttaranchal, Arunachal Pradesh, Meghalaya, Himachal Pradesh (found throughout Himalayas, Kashmir and Khasi Hills of Assam)	Pulmonary affections, dysentery, antiinflammatory, antidiuretic, eye disease, cut and burns, fever, antidiabetic, astringent, cardiotonic, wound healer, antipyretic, hepatoprotective, anticancer, antiprotozoal, antiinflammatory	https://www.bimbima.com/ayurveda/pashanbhedabergenia-ligulata-a-great-herb-to-dissolve-kidney-stone-more/392/; Verma et al. (2014)

Cinchona ledgeriana		Quinine, anthraquinones, purpurin, anthragallol-1,2-dimethylether, anthragallol-1,3-dimethylether, rubiadin, 1-hydroxy-2-hydroxymethylanthr aquinone, 1-hydroxy-2-methylan thraquinone, morindone-5-methylether, anthraquinones	Meghalaya	Antispasmodic, astrigent, antibacterial, antimalarial	Maehara et al. (2013); https://books.google.co.in/books?id=g2Pt45ETJNnkC&lpg=PA297&ots=bv8FvfuJ3i&dq=anthraquinones%20%20in%20natural%20Cinchona%201edgeriana&pg=PA297#v=onepage&q=anthraquinones%20%20in%20natural%20Cinchona% 20ledgeriana&f=false; Wijnsma et al. (1985)
Colchicum luteum Baker		Colchicine, cornigerine, purpurin, anthragallol-1,2-dimethylether, anthragallol-1,3-dimethylether, rubiadin, 1-hydroxy-2-hydroxymethylan thraquinone, 1-hydroxy-2-methylanthraquinone	Ladakh, Kargil, Kashmir to Chamba (Suru valley)	Antigout, stomachache	Singh and Chaurasia (1998); http://www.catalogueoflife.org/annual-checklist/2014/ details/species/id/9766304

Continued

Table 9.1 High-Altitude Medicinal Plants With Their Active Components, Distribution, and Therapeutic Activities—cont'd					
Medicinal Plants	**Images**	**Active Components**	**Distribution**	**Therapeutic Activities**	**References**
Cordyceps sp.		Cordycepin, adenosine, cytosine, thymine, adenine, guanine, uracil, hypoxanthine, uridine-5-mono-phosphate, adenosine-5-monophosphate, guanosine-5-monophosphate	Arunachal Pradesh, Himachal Pradisesh, Uttarakhnnd, Sikkim	Antioxidant, antitumor, anti-inflammatory, anticancer activity, immunomodulatory, antifibrotic, antidiabetic, antiarteriosclerosis, radioprotective activity antithrombotic activity	Rakhee et al. (2016); Mamta et al. (2015)
Curcuma sp.		Curcumin, curcuminoids, 4-hydroxybenzaldehyde	Tripura (Jampui Hill, Suryamaninagar, Baramura hill)	Anticancer activity, cytotoxicity, Bis-demethoxycurcumin	Tripathi et al. (2002), Saha et al. (2016), Kuttan et al. (1985), Nalli et al. (2017)
Ganoderma sp.		Ganoderic acids, riboflavin, ascorbic acid, mannitol, mannitol-alpha, alpha-trehalose, stearic acid, palmitic acid, lignoceric acid, *n*-nonadecanoic acid, behenic acid, tetracosanol, hentriacontane, choline, lycine	Thrissur District, Kerala, Eastern Himalayas, Kashmir valley, Garhwal, South East Maharashtra	Antimutagenic, anticarcinogenic, antiperoxidative activity, antiinflammatory activity	Bhardwaj et al. (2016), Lakshmi et al. (2003), Jones and Kinghorn (2012)

Hippophae sp.		Threonine, valine, methionine, leucine, lysine, trytophan, isoleucine, phenylalanine, ethyl 2-methylbutanoate, ethyl 3-methyl butanoate, alpha-tocopherol	North-eastern Himalayas of India	Cytoprotective, antistress, immunomodulatory, hepatoprotective, radioprotective, antiatherogenic, antitumor, antimicrobial, antiulcerogenic, tissue regeneration	Gao et al. (2000), Geetha et al. (2003)
Nardostachys jatamansi		E-2-methyl, 3-(5,9 dimethylbicyclo[4.3.0] nonen-9-yl)-2-propenoic acid, 2′,2′-dimethyl-3′-methoxy-3′,4′-dihydropyranocoumarin	Uttaranchal, Himachal Pradesh, Arunachal Pradesh, Meghalaya (Grows at heights up to 5000 m in Eastern Himalayas, in Nepal, Bhutan and Sikkim)	Palpitation of heart, intestinal tonic	https://indianbiodiversitytalk.blogspot.in/2012/08/jatamansi-under-threat-in-Himalaya-due-to-illegal-trade.html
Origanum vulgare		P-cymene, γ-terpinene, caryophyllene, spathulenol, germacrene-D, β-fenchyl alcohol, carvacrol, thymol, δ-terpineol	Uttaranchal, Himachal Pradesh (Native to Europe, naturalized in the Middle East)	Volatile oil used as aromatic stimulant in rheumatism	http://powo.science.kew.org/taxon/urn:lsid:ipni.org:names:453395-1; Mockute et al. (2001)
Polygonatum verticilatum		α-Bulnesene, linalyl acetate, eicosadienoic, pentacosane, piperitone, docasane, diosgenin, santonin and calarene	Himachal Pradesh, Uttaranchal, Meghalaya, Arunachal Pradesh (Found in Himalayas at height of 1800 to 3900 m)	Fever, glactagogue, vitiated condition of pitta and vata, aphrodisiac, emollient	Saboon et al. (2016)

Continued

Table 9.1 High-Altitude Medicinal Plants With Their Active Components, Distribution, and Therapeutic Activities—cont'd

Medicinal Plants	Images	Active Components	Distribution	Therapeutic Activities	References
Rhodiola sp.		L-methionine, L-phenylalanine, L-lysine, L-leucine, and L-histidine, palmitoleic acid, oleic acid, linoleic acid	Leh, Ladakh	Adaptogenic, antifatigue, antidepressant, antioxidant, anti-inflammatory, antinoception, anticancer activities, prevent cardiovascular, neuronal, liver, skin disorders	Dhar et al. (2013)
Valeriana wallichii		Boryl isovalerianate, chatinine, formate, glucoside, isovalerenic acid, 1-camphene, 1-pinene, resins, terpincol and valerianine	Uttaranchal, Himachal Pradesh (Distributed throughout drier parts of India)	Alexiteric, cures epileptic seizures, head troubles, disease of eye, blood disease, suppression of urine, astringent carminative, hypnotic, aphrodisiac, pain in joint, disease of liver, clear the voice	Devi and Rao (2014)

Valeriana sp.		Bromazepam, clonazepam, diazepam, valerenic acid, beta-sitosterol, ursolic acid, 4,4′,8,8′-tetrahydroxy-3,3′-dimethoxyl-dibenzyl-ditetrahydrofuran and caryophyllene acide, valerane, naphthalene, linoleic acid, ethyl ester, myrtenyl acetate, ursolic acid, 4,4′,8,8′-tetrahydroxy-3, 3′-dimethoxyl-dibenzyl-ditetrahydrofuran	Himachal Pradesh, it grows profusely in Bharmour division of Chamba, Kanda area of Karsog, and Chansil of Rohru forest division	Treatment of habitual constipation, insomnia, epilepsy, neurosis, anxiety, diuretic, hepatoprotective, analgesic, and cytotoxic, antispasmodic, anticonvulsant, antiinflammatory	Patočka and Jakl (2010), Saklani et al. (2012)
Withania somnifera		Withanolides, withaferin A	Uttaranchal, Himachal Pradesh.	Rheumatism, consumption, coagulating milk, antioxidant, anxiolytic, adaptogen, memory enhancing, antiparkinsonian, antivenom, antiinflammatory, antitumor, immunomodulation, hypolipidemic, antibacterial, cardiovascular protection, sexual behavior	Red Data Book; Wijnsma et al. (1985)

ledgeriana, Colchicum luteum Baker, Cordyceps sp., *Curcuma* sp., *Ganoderma* sp., *Hippophae* sp., *Nardostachys jatamansi, Origanum vulgare, Polygonatum verticilatum, Rhodiola* sp., *Valeriana* sp., and *Withania somnifera* are unique in respect to their biological activities (Balakumbahan et al., 2010; Geetha et al., 2003; Hayes et al., 2008; Kaul, 1997; Mathews et al., 1990; Mishra et al., 2017a, b; Mockute et al., 2001; Patočka and Jakl, 2010; Rakhee et al., 2016; Saboon et al., 2016; Saha et al., 2016; Srimal, 2001; Wijnsma et al., 1985).

Most of the bioactive compounds in the HA plants encompass secondary metabolites (as seen in Table 9.1). For instance, many HA herbal sources consist of substantial amounts of volatile organic compounds (VOCs), which are primarily responsible for imparting aroma and fragrance. Such compounds are used as chief ingredients in essential oils and oil-based healing formulations used to treat arthritis, muscle contractions, and sciatica. Several other secondary metabolites in these herbs are alkaloids, amino acids, fatty acids, flavonoids, glycosides, lignans, nucleotides, nucleoside, phenolic acids, phenols, proteins, resins, sterols, tannins, and terpenoids (Azmir et al., 2013; Ingle et al., 2017).

In light of the abundance of phytoconstituents in herbal sources and their varied bioactivities, it is essential to compile all the analytical techniques that enable isolation and characterization of these compounds. This compilation could further support their application as therapeutic agents.

9.3 EXTRACTION, ISOLATION, AND CHARACTERIZATION OF BIOACTIVE COMPOUNDS FROM HERBAL SOURCES

HA herbal sources have unique salubrious phytoconstituents that offer adaptability to extreme pressure and temperature (Brusotti et al., 2014; Rakhee et al., 2017). A number of advanced and sophisticated techniques that are commonly used to isolate and characterize these bioactive compounds are explained in the following sections (Brusotti et al., 2014; Rakhee et al., 2017).

The extraction method is the first step involved during sample preparation. Two types of extraction methods generally are employed: conventional (boiling under reflux, decoction, infusion, maceration) and nonconventional (microwave-assisted extraction (MAE), ultrasound-assisted extraction (UAE), supercritical fluid extraction (SFE), pressurized-assisted extraction (PAE), hydrotropic extraction, enzyme-assisted extraction, liquid-liquid extraction, liquid phase extraction, solid phase extraction, microporous absorption resin) (Ingle et al., 2017; Sasidharan et al., 2011). The second step focuses on isolation and concentration of the desired natural product using different chromatographic techniques and solidifying methods. The third step includes the analyses of isolated bioactive compounds for chemical characterization (antioxidant assays, enzymatic assays, immunoassays, in vivo and in vitro assays, phytochemical screening, and toxicity evaluation), and genetic characterization (conventional PCR, DNA probe, LAMP, multiplex PCR, SARP, and SSCP-PCR) (Wangchuk, 2014). The isolated herbal products then are subjected to various structure elucidation techniques, including DEPT-20, ESI-MS, FT-IR, ^{1}H NMR, ^{13}C NMR, ^{1}H—^{1}H coupled NMR, 2D-NMR, and MALDI-TOF/TOF (Mishra et al., 2017a, b; Seger et al., 2013; Siddiqui et al., 2017) and, if required, to hyphenated spectroscopic techniques such as HPLC-NMR-MS, HPLC-IR, HPLC-UV-IR, LC-PDA-NMR-MS, LC-UV-MS, LC-TSP-MS, LVI-GC-MS, and SPE-LC-MS (O'Connor and Gibbons, 2013; Potterat and Hamburger, 2013; Sasidharan et al., 2011; Seger et al., 2013; Siddiqui et al., 2017). These techniques have been reported extensively in literature for characterization of biomolecules, even on a nano scale. A schematic representation describing these extraction, isolation, and characterization techniques is given in Fig. 9.2.

9.3.1 Extraction Techniques

Extraction of bioactive compounds, especially secondary metabolites from HA medicinal plants and herbs, depends on several factors, such as polarity and stability of the sample, toxicity, volatility, viscosity, and purity of the extraction solvent (Sasidharan et al., 2011). Several secondary metabolites are present intracellularly so the raw material needs to be broken down or hydrolyzed to improve the extraction yield (Bennett and Wallsgrove, 1994).

9.3.1.1 Conventional Extraction Methods

Some classical and conventional methods of extraction mainly depends on the polarity of solvents and nature of the targeted class of compounds to be isolated (Pereira and Meireles, 2010; Sasidharan et al., 2011).

Distillation

Distillation is the most common method for separating mixtures of secondary metabolites found in medicinal herbs where compounds are separated in different phases (Jones and Kinghorn, 2012). Separation of various bioactive constituents in a mixture is carried out by heating the aqueous form of mixture, wherein different components are separated by their different boiling points, into the gaseous phase. The gas then is condensed back into liquid form and collected. Distillation is used for several purposes, such as production of aromatic components, and phenolic and terpenoid rich constituents. Four types of distillation methods—simple distillation, vacuum distillation, fractional distillation, and steam distillation—are used (Lea and Swoboda, 1962).

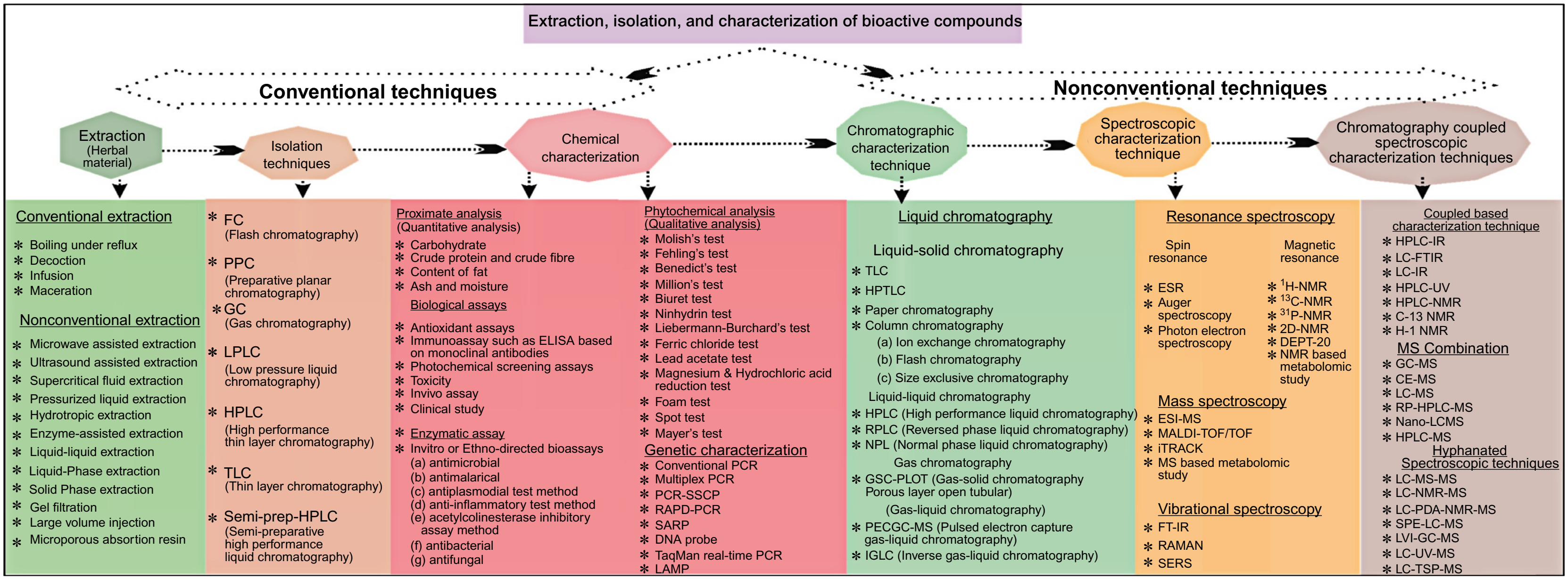


FIG. 9.2

Extraction and characterization techniques of bioactive herbal products.

Maceration

Maceration has been used in domestic preparation of medicinal drugs since ancient times. It gained popularity because of its convenient and cost-effective approach in obtaining essential oils and bioactive compounds. On small-scale extraction, maceration consists of several steps (Jones and Kinghorn, 2012). First, raw material is ground into smaller particles in order to increase the surface area for proper mixing with solvent. Second, an appropriate solvent called "menstruum" is added in a closed vessel to the sample. Third, the liquid is strained off and the "marc, or solid residue, is pressed to recover a large amount of occluded solutions. Infrequent shaking to improve the extraction yield facilitates extraction by increasing diffusion and by removing concentrated solution from the sample surface (Jones and Kinghorn, 2012).

Soxhlet Extraction

Soxhlet extraction has been used widely for extracting valuable bioactive compounds from various natural sources. In this extraction, a small amount of dry sample is placed in a thimble, which is placed in a distillation flask containing the solvent of particular interest. After reaching an overflow level, the solution of the thimble-holder is aspirated by a siphon, which unloads the solution back into the distillation flask. This solution carries the extracted solutes into the bulk liquid. The solute remains behind in the distillation flask, and the solvent passes back to the solid bed of samples. The process is repeated until complete extraction takes place (Saim et al., 1997).

9.3.1.2 Nonconventional Solvent Extraction Methods

Most of the isolation and extraction procedures meant for medicinal plants still use organic solvents of different polarities, water, or their mixtures. Some of the nonconventional extraction methods that are commonly performed are extraction on solid phase, ultrasound-assisted extraction (UAE) (Pingret et al., 2013), extraction with ionic liquids, microwave-assisted extraction (MAE) (Jones and Kinghorn, 2012), accelerated solvent extraction (ASE) (Rakhee et al., 2016), supercritical fluid extraction (SFE) (Paulaitis et al., 1983), and turbo-extraction (Hoberg et al., 1999).

Accelerated Solvent Extraction

ASE is applied to extract compounds of interest from solid and semisolid matrices, using either polar or nonpolar solvents, depending on the type of extract to be prepared. The extraction conditions generally are maintained at an elevated pressure and temperature to maintain the state of the solvent and to achieve enhanced extraction efficiency, respectively. In this extraction method, sequential extraction with solvents of different polarity and mixing of solvents is possible (Rakhee et al., 2016). This mode of extraction has been widely followed to prepare extracts rich in bioactive constituents, including flavonoids, nucleobases, and sterols from numerous herbal sources, such as medicinal mushrooms, sea buckthorn, and *V. wallichii* (Bhardwaj et al., 2015; Mamta et al., 2015; Sharma et al., 2012).

Ionic Liquid Extraction

In recent years, application of ionic liquids for UAE, MAE, or simple batch extraction of plant metabolites at room temperature or elevated temperature has gained increasing attention. Ionic liquids, such as *N*,*N*-dimethylethanolammonium octanoate (DMEA oct), and bis (2 methoxyethyl) ammonium bis (triuoromethylsulfonyl) imide (BMOEA bst) (Cutler et al., 2006; Wu et al., 2015) are used frequently for extraction of natural products from HA medicinal plants and herbs (Qi et al., 2006).

Microwave-Assisted Extraction

Extraction methods employing either diffused microwaves in closed systems or focused microwaves in open systems are being used now for their better efficiency (Jones and Kinghorn, 2012). Many types of microwave-assisted extraction methods are vacuum microwave-assisted extraction (VMAE), nitrogen-protected microwave assisted extraction (NPMAE), ultrasonic microwave-assisted extraction (UMAE), and dynamic microwave-assisted extraction (DMAE) (Ericsson and Colmsjö, 2000; Wang et al., 2008; Yu et al., 2009).

Solid-Phase Extraction

Solid-phase extraction procedures are based on adsorption of analytes or unwanted impurities on a solid phase to aid in purification of natural products. Solid-phase extraction employs a wide range of stationary phases with diverse chemical properties, for example, hydrophilic interaction liquid chromatography (HILIC) stationary phases, ion-exchange resins, mixed-mode material, reversed-phase material, and silica gel (Sasidharan et al., 2011).

Supercritical Fluid Extraction

SFE is a diffusion-based process of separating one component (extractant) from another (matrix) using supercritical fluids. Extraction conditions usually are maintained above the critical pressure (P_c) and critical temperature (T_c) of the extraction solvent in use. The advantages of such an extraction technique is that it offers faster separation, higher selectivity, and increased purity of the extract collected (Villanueva-Bermejo et al., 2017). Water and CO_2 are preferred as extraction solvents for preparing herbal extracts by supercritical fluid extraction because of their ease in handling and nontoxic nature. Additionally, the use of modifiers, such as ethanol in varying percentages, is at times incorporated to accentuate the extraction efficiency by alteration of polarities (Paulaitis et al., 1983). SFE recently has been proven to be highly beneficial in preparing bioactive extracts from plant sources. For example, extracts prepared from leaves of *Hippophae rhamnoides* using supercritical CO_2 have shown appreciable adjuvant activities against *diphtheria* and *tetanus* toxoids (Jayashankar et al., 2017).

Ultrasound-Assisted Extraction

In UAE, the plant material is placed in a glass container, covered with the appropriate extraction solvent, and put into an ultrasonic bath. This extraction technique involves considerable mechanical stress that induces cavitation and cellular breakdown, resulting in an increased extraction yield. The extraction time also is substantially reduced (Pingret et al., 2013).

9.3.1.3 Specialized Extraction Methods

Apart from the conventional and nonconventional extraction methods, a few specialized techniques also are adopted.

Preparative High-Performance Liquid Chromatography

Preparative high-performance liquid chromatography generally is used to obtain a high yield of analytes at a good resolution. This technique is appropriate for characterizing samples belonging to different classes of bioactive compounds, such as flavonoids, phenolic compounds, and vitamins. The different chemical components in the samples are fractionated according to their retention times. The fractions are subjected to further characterization studies such as antioxidant assays, biochemical analyses, and other miscellaneous biological assays (Aravind et al., 2008).

Preparative Planar Chromatography

Preparative planar chromatography is a type of liquid chromatography used to isolate bioactive compounds in a small sample size (range 1–100 mg). This technique is typically advantageous to purify small amounts of samples to be used in preparative and isolation purposes (Hostettmann et al., 1986).

9.3.2 Solvent Removal Techniques for the Preparation of Herbal Extracts

After carrying out any of the extraction methods, the sample extract is introduced to an appropriate drying process. The dried sample extract then is recovered by means of several concentration procedures while discarding residual solvent from the prepared extract (Adou, 2005). The concentrated solid extract is subjected to appropriate chromatographic techniques in order to identify specific and nonspecific types of phytoconstituents (Sasidharan et al., 2011).

9.3.2.1 Solvent Evaporation Techniques

Some of the frequently applied techniques to dry the extracted sample for solidification, purification, and concentration use hot air, microwave, and oven along with a freeze-drying method to purify and concentrate the sample (Saim et al., 1997).

Drying by Conventional Techniques

The extracted sample bearing high medicinal value is subjected to conventional methods of drying by use of hot air, microwave, or oven to increase its concentration and also to aid in removal of any residual impurities. The temperature in this process is controlled to ensure that no chemical decomposition takes place in the sample while the dried product is retrieved.

Lyophilization

Freeze drying or lyophilization of extracted samples from medicinal plants is used to remove solvents (usually water) through a process called sublimation. In sublimation, the solid frozen material is converted directly to a gaseous phase without passing through a liquid phase (Wang, 2000). For this, samples are frozen, and all the components are precooled to attain a frozen state. The ice then is sublimed in two steps. In the primary drying phase, about 98% of the water gets sublimed; the secondary drying phase aims to sublime the bound water molecules, resulting in a final purified extracted sample (Wang, 2000).

Rotary Vaporization

This type of drying is well-suited for the concentration of samples containing primary and secondary plant metabolites, which could be separated by evaporation (Mukherji et al., 2013). In rotary vaporization, the extracted sample is evaporated at a certain temperature under vacuum to remove the solvent from the sample, thereby leaving behind a dried, solvent-free concentrated sample. The dried extract is subsequently dissolved in deionized water and defatted with petroleum ether to remove fats. If required, successive partitioning with solvents of different polarities is carried out using a separating funnel to obtain a more purified sample (Mukherji et al., 2013).

Centrifugal Evaporation

This type of drying is achieved by speed-vacuum concentrator, where organic solvents such as alcohols, ethers, and chloroform, are dried under vacuum, i.e., low pressure and ambient temperatures. This facilitates shifting of the vapor-liquid equilibrium of the solvent toward the gas phase, while the sample being dried remains in solid phase. This is a gentler mode of drying than rotary vaporization, where drying occurs under elevated temperatures (Roth et al., 2010).

9.3.3 Characterization Techniques for Bioactive Constituents

The characterization methods to be adopted for any isolated herbal extract depends on the nature of the extract and on the type of extraction techniques that were followed to prepare the extract. Characterization largely falls under two categories: conventional and nonconventional techniques.

9.3.3.1 Conventional Characterization Techniques

Many chemical methods are used for qualitative evaluation of several phytochemicals, such as alkaloids, amino acids, fat and oils, glycosides, proteins, saponins, steroids and sterols, sugars, tannins, and terpenoids (Rakhee et al., 2016). Some of the specific qualitative tests that have been used for the primary level characterization of the pharmacologically active phytoconstituents are described below (Rakhee et al., 2017).

Proximate Analysis

HA herbs have gained widespread attention because of their richness in numerous types of phytoconstituents bearing pharmacological and nutraceutical properties. Proximate analysis is the most preferred analytical technique for evaluating the quantity of amino acids, ash, carbohydrates, fats, minerals, moisture, and protein contents present in extracts of HA medicinal plants (Rakhee et al., 2016).

Biochemical Characterization

Extracts prepared from HA medicinal herbs are subjected to estimation of biochemical activity in terms of standard curves wherein the standards comprise the desired class of bioactive compound under study. A number of biochemical assays such as DPPH (2,2-diphenyl-1-picryhydrazyl), $ABTS^{\cdot +}$ (2,2′-azino-bis (3-ethylbenzothiazoline-6-sulphonic acid), FRAP (ferric-reducing antioxidant power), MDA (malondialdehyde), SOD (superoxide dismutase), and biological assays aid in the estimation of antibacterial activity, antiproliferative activity, and protein leakage (Geetha et al., 2003; Mishra et al., 2017a, b; Sharma et al., 2012). These assays have established the importance of HA plants in terms of antimicrobial, antioxidant, antitumor, immunomodulating, and other beneficial activities (Geetha et al., 2005; Mishra et al., 2017a, b).

9.3.3.2 Nonconventional Characterization Techniques

Nonconventional characterization techniques include relatively advanced and current techniques that are being implemented worldwide for the characterization of phytochemical constituents.

Chromatographic Characterization Techniques

Identification of secondary metabolites in herbal products usually is a tedious process because of their complex compositions. Nevertheless, a couple of chromatography-based techniques enable easy identification of metabolites, thereby facilitating characterization of any type of bioactive compound. Some chromatographic-based techniques that are generally adopted are described in this section.

Thin-Layer Chromatography. Thin-layer chromatography (TLC) is one of the most popular and simplistic chromatographic techniques used for separation of compounds. TLC is employed extensively in phytochemical evaluation of herbal drugs because of its advantages, such as rapid analysis of herbal extracts, minimum sample clean up requirement, and ease in acquiring qualitative and quantitative information of the resolved compounds (Siddiqui et al., 2017).

High-Performance Thin-Layer Chromatography. High-performance thin-layer chromatography (HPTLC) is employed widely in pharmaceutical industries for process development, identification, and detection of adulterants in herbal products; identification of pesticide content and mycotoxins; and in quality control of herbs and health food. HPTLC has been reported to offer several advantages over the more widely used high-performance liquid chromatography (HPLC) technique because of its high throughput (multiple samples can be analyzed simultaneously), economy in use of mobile phase, and cost effectiveness (Mishra et al., 2017a, b).

High-Performance Liquid Chromatography. Over the past decades, HPLC has become the most expansively applied analytical technique for the analysis of herbal medicines. Usually, reversed phase (RP) columns are used for this purpose. Both analytical and preparative modes of HPLC are used in pharmaceutical industries for isolation and purification of herbal compounds. The parameters that need to be considered while performing characterization by HPLC are rapid analysis time, resolution, sensitivity, and specificity (Yan et al., 2007).

Low-Pressure Liquid Chromatography. Column chromatographic methods that allow the use of mobile phase at atmospheric pressure, without applying additional forces, are used as a major tool in the fractionation protocols followed for isolation of natural products. This technique uses hydroxypropylated dextran gel Sephadex LH-20 that is based not only on principle of molecular sieves, but also on separation by adsorption (Scalbert et al., 1990).

Gas Chromatography. Gas chromatography (GC) is a well-established analytical technique that offers high resolution and sensitivity, particularly in the characterization, quantification, and identification of volatile organic compounds (VOCs). Thus, GC is used mainly in the industries dealing with cosmetics, environmental toxins, and pharmaceuticals (Chauhan, 2014). Essential oils, fatty acids, and other secretions present in herbal extracts can be analyzed easily using GC (Ashley et al., 1996).

Flash Chromatography. Flash chromatography (FC) is an advanced type of chromatography that is used for rapid separation of aromatic, organic, or phenolic compounds. This is an air pressure driven technique comprising medium and short column chromatography (Hostettmann et al., 1986).

Spectroscopic Techniques

Ultraviolet-Visible Spectroscopy. Ultraviolet-visible spectroscopy (UV-Vis spectroscopy) makes use of absorption spectroscopy in ultraviolet and visible wavelength ranges—180–380 nm and 380–750 nm, respectively—for characterizing molecules. This is one of the most basic techniques that need to be conducted while characterizing an analyte. All the major classes of biomolecules contain certain light absorbing functional groups known as chromophores. Upon absorbing UV/Vis light, these chromophores get excited from ground state to a higher energy level, thus giving out characteristic spectra, aiding in the identification of specific biomolecules (Perkampus and Grinter, 1992).

Fourier Transform-Infrared Spectroscopy. Fourier transform-infrared spectroscopy (FT-IR) is based on the fact that most molecules absorb light in the infrared region of the electromagnetic spectrum. The frequency ranges are measured as wave numbers typically over the range 4000–400 cm^{-1}. FT-IR is useful mainly to identify organic molecular groups and compounds because of the range of functional groups. Pellets of samples are prepared with KBr salts owing to their transparency, thus offering better resolution of the spectrum (Faix, 1991). FT-IR is the most commonly used technique for structure elucidation of bioactive herbal extracts. For example, the IR spectrum of the leaf extract of *Euphorbia thymifolia* from the Kumaon Himalayas exhibited absorption band at 3407 cm^{-1} for hydroxyl group, 1666 cm^{-1} for carbonyl group, and 1562 cm^{-1} for unsaturation (Prasad, 2008).

Electrospray Ionization-Mass Spectrometry. Electrospray ionization-mass spectrometry (ESI-MS) generates ions upon application of an electric potential to a flowing liquid, causing the liquid to get charged and then sprayed. This electrospray forms very small droplets of solvent containing the analytes. ESI permits the coupling of HPLC instruments directly with MS. ESI-MS technique is used to characterize phenolic-rich fractions of medicinal plants as well as for identification of flavonoids such as gentenstein 7-glucoside, iridin, orientin, dihexosyl quercetin, quercetin-3-O-rutinoside, rhamnosyl hexosyl methyl quercetin, C-(O-caffeoyl-hexosyl)-O-hexoside, and tricin 7-O-hexosyl-O-hexoside in *Tragia involucrate* (Sulaiman and Balachandran, 2016).

Nuclear Magnetic Resonance Spectroscopy

Proton NMR Spectroscopy. Proton nuclear magnetic resonance (^{1}H NMR) spectroscopy is used to identify different types of protons having a nuclear spin $(I)=1/2\,m$, in herbal samples. This technique shows the presence of magnetically induced shielded and deshielded protons in terms of characteristic coupling constant (J) (Berger and Braun 2004; Pavia et al., 2008).

Carbon-13 NMR Spectroscopy. This spectroscopic technique is based on the magnetically active nuclei ^{13}C, which has very low natural abundance (approximately, 1.1%) in comparison to ^{12}C. The spectrum generated here is recorded as a proton decoupled form as singlet and gives an idea about the types of carbon atoms having different types of environment. The ^{13}C NMR delta values normally range between 0 and 220 ppm. Typically, tetramethylsilane is used as the internal reference compound and samples are prepared in d^{6}-DMSO (Pavia et al., 2008; Seger et al., 2013).

Two-Dimensional NMR Spectroscopy (2D-NMR). Complex overlapping signals in NMR spectra could be interpreted easily by two main strategies. The first strategy aims at simplifying the ^{1}H NMR spectrum through creation of a projection of the ^{1}H broad-band decoupled spectrum as in J-resolved experiment spectroscopy NMR (JRES-NMR), or through the selective suppression of signals from certain compounds using relaxation or diffusion filters as in diffusion ordered-NMR spectroscopy (DOSY) (Mahrous and Farag, 2015; Pavia et al., 2008). While both JRES and DOSY are 2D techniques, the most useful aspect in both the methods is the ability to extract simplified proton spectrum with less peak interference rather than collecting information displayed in the second dimension, such as coupling constant and translational diffusion coefficient, respectively (Pavia et al., 2008).

Matrix-Assisted Laser Desorption/Ionization-Time of Flight Mass Spectrometry (MALDI-TOF/TOF). MALDI is a soft ionization technique for the characterization of different bioactive peptides and proteins. The sample to be analyzed by MALDI is uniformly mixed with a large quantity of matrix. The matrix absorbs nitrogen laser light of wavelength 337 nm and converts it into heat energy. A small part of the matrix gets heated rapidly and is vaporized, along with the sample (Rakhee et al., 2016; Rakhee et al., 2017).

Hyphenated Techniques

Recent advances in characterization techniques have witnessed the hyphenation of spectroscopic techniques with regular and/or complex types of chromatography such as GC, HPLC, HPTLC, and ion-exchange chromatography. Some are described below.

High-Pressure Liquid Chromatography With UV Detector. High-pressure liquid chromatography with UV detector (HPLC-UV) technique is commonly used to isolate and characterize medicinally important compounds found in many HA herbal sources, such as rosarin and rosin. In one study, eight different chemotaxonomic markers, namely, 4-(+) catechin, gallic acid, 8-rhodionin, 5-rosarin, 6-rosavin, 7-rosin, 3-salidroside, and 2-tyrosol were characterized in different *Rhodiola* species using HPLC-DAD/UV technique (Liu et al., 2013).

High-Pressure Liquid Chromatography-Mass Spectrometry. HPLC with diode array detector coupled with ESI-MS (HPLC-DAD-ESI-MS/MS) has been well-adopted for the characterization of many natural products. This technique has been employed for simultaneous identification of flavonoids and other constituents in the **rhizomes** of three *Iris* species growing in Kashmir valleys of India (Bhat, 2014; Seger et al., 2013). About 27 compounds were identified in these *Iris* species based on UV spectra, MS/MS spectra and retention time. Among those 27 compounds, seven constituents from *Iris crocea* (major being hesperetin, iridal glycoside 5b, iridal glycoside 7, irilin A, irisolodone, irisolone and irisquin B) were reported (Bhat, 2014).

High-Pressure Liquid Chromatography-Nuclear Magnetic Resonance. NMR spectroscopy is by far the most powerful spectroscopic technique for obtaining structural information about organic compounds in solution. The direct coupling of LC with proton NMR has been reported in many papers. Early experiments of coupled HPLC-^{1}H NMR have been conducted in a stop-flow mode or with very minimal flow rates. This type of flow is necessary to accumulate a sufficient number of spectra per sample volume in order to improve the S/N ratio (Seger et al., 2013). One example of application of this technique is isolation of iridoid glycosides *in Picrorhiza kurroa* Royle, an important medicinal plant thriving in HA. Iridoid glycosides were isolated on a RP C-18 column using a mobile phase of methanol and water (40:60 *v*/v). Further characterization of these glycosides was performed by ^{13}C NMR, 2D-NMR, ^{1}H NMR, MS, and UV. These iridoid glycosides, namely, picrosides I, II and kutkoside, are used in various herbal drug formulations as strong hepatoprotective and immunomodulatory compounds (Sultan et al., 2016).

Liquid Chromatography-Mass Spectrometry. Liquid chromatography-mass spectrometry (LC-MS) has undergone tremendous technological improvements, thus permitting its application to analyze endogenous metabolites, such as carbohydrates, DNA, drugs, peptides, and proteins (Kang, 2012). This method has been used to characterize many bioactive components occurring in HA medicinal plants, such as in UAE-yielded extracts of *Actinocarya tibetica*, five bioactive flavonoids—gardenin, 5-methoxy-6,7-methylenedioxyflavone, mosloflavone, negletein, and 5,6,7-trimethoxyflavone (Balakumbahan et al., 2010; Singh et al., 2013).

Gas Chromatography-Mass Spectrometry. Gas chromatography-mass spectrometry (GC-MS) is a hyphenated analytical technique that combines the separation properties of GC with the detection feature of MS to identify different substances within a test sample (Bhardwaj et al., 2017). A large number of volatile organic compounds and essential oils of high medicinal value have been isolated by liquid chromatography or gas chromatography and further characterized using FTIR or MS techniques. In different varieties of the *Valeriana* genus, GC and LC have been applied effectively along with MS to confirm the presence of E-valerenyl isovalerate, furinol-1, furinol-2, hydroxyvalerenic acid, methyl valiant, pacifigorgiol, sesquiterpenoids, valerenic acid, valerenol, and *Z*-Valerenyl acetate (Bos, 1997; Chauhan, 2014).

Reversed Phase-High-Performance Liquid Chromatographic-Mass Spectrometry. Reversed phase-high-performance liquid chromatographic-mass spectrometry (RP-HPLC-MS) involves the separation of molecules based on hydrophobicity. Separation depends on the hydrophobic binding of the solute molecule from the mobile phase to the immobilized hydrophobic ligands attached to the stationary phase, the sorbent. In different HA medicinal plants such as *Hippophae rhamnoides*, *Prunus armeniaca*, and *Rhodiola imbricate*, RP-HPLC tagged with precolumn derivatization and CE have been used identify and quantify amino acids and vitamins (E and B-complex vitamins) (Xiong et al., 2013). RP-HPLC-MS has also been used to characterize essential and nonessential amino acids such as L-methionine, L-phenylalanine, L-lysine, L-leucine, and L-histidine in herbal extracts (Xiong et al., 2013).

Thin-Layer Chromatography-Mass Spectrometry. Advancement in TLC has witnessed its hyphenation with mass spectrometry for confirmation of biomolecules. In this technique, a particular band (corresponding to the analyte under study) is extracted from the TLC or HPTLC plate by an appropriate interface and is subjected to mass spectrometry to ensure better identification. This technique is particularly useful for resolving the kind of analytes that might not be distinguished because of closer retardation factor values but could be differentiated based on their varying m/z values. TLC-MS has been applied to identify phytoconstituents in herbal samples. In a recent study, TLC-MS aided in identification of nucleobases in *Hippophae* species extracts (Mishra et al., 2017a, b).

9.4 MULTIHYPHENATED CHARACTERIZATION TECHNIQUES

Hyphenation of spectroscopic and chromatographic techniques is not limited to two or three techniques. The coupling could involve more than one separation or detection techniques, such as LC-PDA-MS, LC-MS-MS, LC-NMR-MS, and LC-PDA-NMR-MS (Patel et al., 2010). The two key elements in natural product research are the isolation and purification of compounds present in crude extracts or fractions obtained from various natural sources, and the unambiguous identification of the isolated compounds (Patel et al., 2010). Thus, the online characterization of secondary metabolites in crude natural product extracts or fractions demands a high degree of sophistication and richness of structural information, sensitivity, and selectivity. The development of various hyphenated techniques has provided natural product researchers with extremely powerful tools that facilitate excellent separation efficiency as well as acquisition of online complimentary spectroscopic data on a GC or LC peak of interest within a complex mixture (Patel et al., 2010).

9.5 CONCLUSIONS

This chapter has described all the relevant extraction, solvent removal, and characterization techniques that are necessary to prepare therapeutic extracts from herbal sources. It could be well ascertained that both conventional as well as nonconventional techniques complement each other for facilitating complete characterization of any herbal extract.

Acknowledgments

The authors are thankful to the director, DIPAS, for providing facilities, support, and encouragement. Ms. Rakhee is grateful to dead, Department of Chemistry, University of Delhi, Delhi, India (09/045 (1463/2017-EMR-I)), for guidance and support. She also would like to acknowledge CSIR for her fellowship grant.

References

Adou, E., 2005. Isolation and Characterization of Bioactive Compounds from Suriname and Madagascar Flora. II. A Synthetic Approach to Lucilactaene. PhD Thesis, Blacksburg, Virginia. URI: http://hdl.handle.net/10919/29973.

Aravind, S., Arimboor, R., Rangan, M., Madhavan, S.N., Arumughan, C., 2008. Semipreparative HPLC preparation and HPTLC quantification of tetrahydroamentoflavone as marker in Semecarpus anacardium and its polyherbal formulations. J. Pharm. Biomed. Anal. 48 (3), 808–813.

Ashley, D.L., Bonin, M.A., Cardinal, F.L., Mccraw, J.M., Wooten, J.V., 1996. Measurement of volatile organic compounds in human blood. Environ. Health Perspect. 104 (Suppl 5), 871–877.

Azmir, J., Zaidul, I.S.M., Rahman, M.M., Sharif, K.M., Mohamed, A., Sahena, F., Jahurul, M.H.A., Ghafoor, K., Norulaini, N.A.N., Omar, A.K.M., 2013. Techniques for extraction of bioactive compounds from plant materials: A review. J. Food Eng. 117 (4), 426–436.

Balakumbahan, R., Rajamani, K., Kumanan, K., 2010. Acorus calamus: an overview. J. Med. Plants Res. 4 (25), 2740–2745.

Bennett, R., Wallsgrove, R., 1994. Secondary metabolites in plant defence mechanisms. New Phytol. 127 (4), 617–633.

Berger, S., Braun, S., 2004. 200 and More NMR Experiments. Wiley-VCH, Weinheim, pp. 305–307.

Bhardwaj, A., Gupta, P., Kumar, N., Mishra, J., Kumar, A., Rakhee, Misra, K., 2017. Lingzhi or Reishi medicinal mushroom, Ganoderma lucidum (Agaricomycetes), inhibits candida biofilms: a metabolomic approach. Int. J. Med. Mushrooms 19 (8), 685–696.

Bhardwaj, A., Pal, M., Srivastava, M., Tulsawani, R., Sugadev, R., Misra, K., 2015. HPTLC based chemometrics of medicinal mushrooms. J. Liq. Chromatogr. Relat. Technol. 38 (14), 1392–1406.

Bhardwaj, A., Srivastava, M., Pal, M., Sharma, Y.K., Bhattacharya, S., Tulsawani, R., Sugadev, R., Misra, K., 2016. Screening of Indian lingzhi or reishi medicinal mushroom, Ganoderma lucidum (agaricomycetes): a UPC2-MS approach. Int. J. Med. Mushrooms 18 (2), 177–189.

Bhat, G., 2014. HPLC-DAD-ESI-MS/MS identification and characterization of major constituents of Iris crocea, Iris germanica, and Iris spuria growing in Kashmir Himalayas, India. J. Anal. Bioanal. Tech. 5 (6), 1–10.

Bos, R., 1997. Analytical and Phytochemical Studies on Valerian and Valerian-Based Preparations (Dissertation). State University of Groningen, Department of Pharmaceutical Biology, Groningen, Netherlands, pp. 184–193.

Brusotti, G., Cesari, I., Dentamaro, A., Caccialanza, G. and Massolini, G., 2014. Isolation and characterization of bioactive compounds from plant resources: The role of analysis in the ethnopharmacological approach. J. Pharm. Biomed. Anal., 87, pp.218–228. DOI: https://doi.org/10.1016/j.jpba.2013.03.007.

Bucar, F., Wube, A., Schmid, M., 2013. Natural product isolation—how to get from biological material to pure compounds. Nat. Prod. Rep. 30 (4), 525–545.

C T, S., Balachandran, I., 2016. LC/MS characterization of antioxidant flavonoids from Tragia involucrata L. Beni-Suef Univ. J. Basic Appl. Sci. 5 (3), 231–235.

Chauhan, A., 2014. GC-MS technique and its analytical applications in science and technology. J. Anal. Bioanal. Tech. 5 (6), 1–5.

Chen, F., Tholl, D., D'Auria, J.C., Farooq, A., Pichersky, E., Gershenzon, J., 2003. Biosynthesis and emission of terpenoid volatiles from Arabidopsis flowers. Plant Cell 15 (2), 481–494.

Cutler, M., Lapkin, A., Plucinski, P.K., 2006. Comparative Assessment of Technologies for Extraction of Artemisinin. Summary Report Commissioned Through the Malaria Medicines Venture. Geneva, August.

Devi, V.S., Rao, M.G., 2014. Valeriana wallichii-A rich aroma root plant. World J. Pharm. Pharm. Sci. 3, 1516–1525.

Dhar, P., Tayade, A.B., Kumar, J., Chaurasia, O.P., Srivastava, R.B., Singh, S.B., 2013. Nutritional profile of Phytococktail from trans-Himalayan plants. PLoS ONE. 8 (12). e83008.

Ericsson, M., Colmsjö, A., 2000. Dynamic microwave-assisted extraction. J. Chromatogr. A 877 (1), 141–151.

Faix, O., 1991. Classification of lignins from different botanical origins by FT-IR spectroscopy. Holzforschung Int. J. Biol. Chem. Phys. Technol. Wood 45 (s1), 21–28.

Gao, X., Ohlander, M., Jeppsson, N., Björk, L., Trajkovski, V., 2000. Changes in antioxidant effects and their relationship to phytonutrients in fruits of sea buckthorn (*Hippophae rhamnoides* L.) during maturation. J. Agric. Food Chem. 48 (5), 1485–1490.

Geetha, S., Ram, M.S., Mongia, S., Singh, V., Ilavazhagan, G., Sawhney, R., 2003. Evaluation of antioxidant activity of leaf extract of Seabuckthorn (Hippophae rhamnoides L.) on chromium (VI) induced oxidative stress in albino rats. J. Ethnopharmacol. 87 (2), 247–251.

Geetha, S., Singh, V., Ram, M.S., Ilavazhagan, G., Banerjee, P., Sawhney, R., 2005. Immunomodulatory effects of sea buckthorn (Hippophae rhamnoides L.) against chromium (VI) induced immunosuppression. Mol. Cell. Biochem. 278 (1), 101–109.

Gođevac, D., Jadranin, M., Aljančić, I., Vajs, V., Tešević, V., Milosavljević, S., 2015. Application of Spectroscopic Methods and Hyphenated Techniques to the Analysis of Complex Plant Extracts. Vol. 1. Springer, Budapest, Hungary, pp. 1–459.

Gupta, P., Dutta, B., Pant, D., Joshi, P., Lohar, D., 2009. In vitro antibacterial activity of *Artemisia annua* Linn. growing in India. Int. J. Green Pharm. 3 (3), 255.

Hayes, P.Y., Jahidin, A.H., Lehmann, R., Penman, K., Kitching, W., De Voss, J.J., 2008. Steroidal saponins from the roots of Asparagus racemosus. Phytochemistry 69 (3), 796–804.

Hoberg, E., Orjala, J., Meier, B., Sticher, O., 1999. Diterpenoids from the fruits of Vitex agnus-castus. Phytochemistry 52 (8), 1555–1558.

Hostettmann, K., Marston, A., Hostettmann, M., 1986. Preparative Chromatography Techniques, second ed. Springer, Berlin, pp. 1–247.

Ingle, K.P., Deshmukh, A.G., Padole, D.A., Dudhare, M.S., Moharil, M.P., Khelurkar, V.C., 2017. Phytochemicals: extraction methods, identification, and detection of bioactive compounds from plant extracts. J. Pharmacogn. Phytochem. 6 (1), 32–36.

Jayashankar, B., Singh, D., Tanwar, H., Mishra, K.P., Murthy, S., Chanda, S., Mishra, J., Tulswani, R., Misra, K., Singh, S.B., Ganju, L., 2017. Augmentation of humoral and cellular immunity in response to Tetanus and Diphtheria toxoids by supercritical carbon dioxide extracts of *Hippophae rhamnoides* L. leaves. Int Immunopharmacol. 44, 123–136.

Jones W.P., Kinghorn A.D. (2012) Extraction of plant secondary metabolites. In: Sarker S., Nahar L. (eds Online ISBN: 978–1–61779-624-1), Natural Products Isolation. Methods in Molecular Biology (Methods and Protocols), Vol. 864. Publisher: Humana Press. DOI:https://doi.org/10.1007/978-1-61779-624-1_13.

Kale, A., Gawande, S., Kotwal, S., 2008. Cancer phytotherapeutics: role for flavonoids at the cellular level. Phytother. Res. 22 (5), 567–577.

Kang, J.S., 2012. Principles and applications of LC-MS/MS for the quantitative bioanalysis of analytes in various biological samples, Tandem Mass Spectrometry-Applications and Principles. Jeevan Prasain (Ed.), ISBN: 978–953–51-0141-3, Seoul, South Korea, Publisher: InTech, Available from: http://www.intechopen.com/books/tandem-massspectrometry-applications-and-principles/principles-and-applications-of-lc-ms-ms-for-the-quantitativebioanalysis-of-analytes-in-various-biol.

Kaul, M., 2010. High Altitude Botanicals in Integrative Medicine—Case Studies From Northwest Himalaya. ISSN: 0975–1068 (Online); 0972–5938 (Print), Vol. 09(1), Publisher: CSIR. pp.18–25. URL: http://hdl.handle.net/123456789/7149.

Kaul, M.K., 1997. Medicinal plants of Kashmir and Ladakh: Temperate and Cold Arid Himalaya. ISBN: 8173870616, 9788173870613,New Delhi, Publisher: Indus publishing, pp. 1–173.

Kuttan, R., Bhanumathy, P., Nirmala, K., George, M., 1985. Potential anticancer activity of turmeric (*Curcuma longa*). Cancer Lett. 29 (2), 197–202.

Lakshmi, B., Ajith, T., Sheena, N., Gunapalan, N., Janardhanan, K., 2003. Antiperoxidative, anti-inflammatory, and antimutagenic activities of ethanol extract of the mycelium of *Ganoderma lucidum* occurring in South India. Teratog. Carcinog. Mutagen. 23 (S1), 85–97.

Lea, C., Swoboda, P., 1962. Simple vacuum distillation procedure for determination of the volatile carbonyl content of autoxidising edible fats. J. Sci. Food Agric. 13 (3), 148–158.

Liu, Z., Liu, Y., Liu, C., Song, Z., Li, Q., Zha, Q., Lu, C., Wang, C., Ning, Z., Zhang, Y., 2013. The chemotaxonomic classification of Rhodiola plants and its correlation with morphological characteristics and genetic taxonomy. Chem. Central J. 7 (1), 1–118.

Maehara, S., Simanjuntak, P., Maetani, Y., Kitamura, C., Ohashi, K., Shibuya, H., 2013. Ability of endophytic filamentous fungi associated with *Cinchona ledgeriana* to produce Cinchona alkaloids. J. Nat. Med. 67 (2), 421–423.

Mahrous, E.A., Farag, M.A., 2015. Two-dimensional NMR spectroscopic approaches for exploring plant metabolome: a review. J. Adv. Res. 6 (1), 3–15.

Mamta, Mehrotra, S., Kirar, V., Vats, P., Nandi, S.P., Negi, P., Misra, K., 2015. Phytochemical and antimicrobial activities of Himalayan *Cordyceps sinensis* (Berk.) Sacc. Indian J Exp Biol. 53 (2015), 36–43.

Mathews, H., Bharathan, N., Litz, R., Narayanan, K., Rao, P., Bhatia, C., 1990. The promotion of Agrobacterium mediated transformation in Atropa belladona L. by acetosyringone. J. Plant Physiol. 136 (4), 404–409.

Mishra, J., Hande, P., Sharma, P., Bhardwaj, A., Rajput, R., Misra, K., 2017. Characterization of nucleobases in sea buckthorn leaves: an HPTLC approach. J. Liquid Chromatogr. Related Technol. 40 (1), 50–57.

Mishra, J., Rajput, R., Singh, K., Puri, S., Goyal, M., Bansal, A. and Misra, K., 2017. Antibacterial natural peptide fractions from Indian Ganoderma lucidum. Int. J. Pept. Res. Ther., pp. 1–12. https://doi.org/10.1007/s10989-017-9643-z

Mockute, D., Bernotiene, G., Judzentiene, A., 2001. The essential oil of Origanum vulgare L. ssp. vulgare growing wild in Vilnius district (Lithuania). Phytochemistry 57 (1), 65–69.

Mukherji, R., Joshi-Navare, K., Prabhune, A., 2013. Crystalline xylitol production by a novel yeast, Pichia caribbica (HQ222812), and its application for quorum sensing inhibition in gram-negative marker strain Chromobacterium violaceum CV026. Appl. Biochem. Biotechnol. 169 (2013), 1753–1763.

Nalli, M., Ortar, G., Moriello, A.S., Di Marzo, V., De Petrocellis, L., 2017. Effects of curcumin and curcumin analogues on TRP channels. Fitoterapia 122, 126–131.

O'Connor, S.E. and Gibbons, S., 2013. Editorial: Modern methods in plant natural products themed issue. Nat. Prod. Rep., 30(2013), pp.483–484. DOI: 10.1039/C3NP90008H.

Patel, K.N., Patel, J.K., Patel, M.P., Rajput, G.C., Patel, H.A., 2010. Introduction to hyphenated techniques and their applications in pharmacy. Pharm. Methods 1 (1), 2–13.

Patočka, J., Jakl, J., 2010. Biomedically relevant chemical constituents of Valeriana officinalis. J. Appl. Biomed. 8 (1), 11–18.

Paulaitis, M.E., Krukonis, V.J., Kurnik, R.T., Reid, R.C., 1983. Supercritical fluid extraction. Rev. Chem. Eng. 1 (2), 179–250.

Pavia, D.L., Lampman, G.M., Kriz, G.S. and Vyvyan, J.A., 2008. Introduction to Spectroscopy. ISBN: 9781111800628, Publisher: Cengage Learning, pp. 1–752.

Pereira, C.G., Meireles, M.A.A., 2010. Supercritical fluid extraction of bioactive compounds: fundamentals, applications, and economic perspectives. Food Bioprocess Technol. 3 (3), 340–372.

Perkampus, H.H., Grinter, H.C., 1992. UV-VIS Spectroscopy and its Applications ISBN: 978-3-642-77479-9(print), 978-3-642-77477-5(online). Publisher: SpringerBerlin-Heidelberg, pp.1–244. https://doi.org/10.1007/978-3-642-77477-5

Pingret, D., Fabiano-Tixier, A.S., Chemat, F., 2013. Natural product extraction: principles and applications. In: Mauricio, R.A., Juliana, P.M. (Eds.), Ultrasound-Assisted Extraction. RSC Publishing, pp. 89–112.

Potterat, O., Hamburger, M., 2013. Concepts and technologies for tracking bioactive compounds in natural product extracts: generation of libraries, and hyphenation of analytical processes with bioassays. Nat. Prod. Rep. 30 (4), 546–564.

Prasad, K., 2008. Phytochemical Investigation of Euphorbia, Pouzolzia, and Pavetta species from Kumaon Himalayas. PhD thesis, Department of Chemistry, Kumaun University, pp. 1–144.

Qi, S., Li, Y., Deng, Y., Cheng, Y., Chen, X., Hu, Z., 2006. Simultaneous determination of bioactive flavone derivatives in Chinese herb extraction by capillary electrophoresis used different electrolyte systems-borate and ionic liquids. J. Chromatogr. A 1109 (2), 300–306.

Rakhee, Sethy, N.K., Bhardwaj, A., Singh, V.K., Sharma, R.K., Deswal, R., Bhargava, K., Misra, K., 2017. Characterization of Ganoderma lucidum: phytochemical and proteomic approach. J. Proteins Proteom. 8 (1), 25–33.

Rakhee, Sethy, N.K., Singh, V.K., Sharma, S., Sharma, R.K., Deswal, R., Bhargava, K., Misra, K., 2016. Phytochemical and proteomic analysis of a high altitude medicinal mushroom Cordyceps sinensis. J. Proteins Proteom. 7 (3), 187–197.

Roth, S., Feichtinger, J., Hertel, C., 2010. Characterization of Bacillus subtilis spore inactivation in low-pressure, low-temperature gas plasma sterilization processes. J. Appl. Microbiol. 108 (2), 521–531.

Saboon, Bibi, Y., Arshad, M., Sabir, S., Amjad, M.S., Ahmed, E., Chaudhari, S.K., 2016. Pharmacology and biochemistry of Polygonatum verticillatum: a review. J. Coast. Life Med. 4 (5), 406–415.

Saha, L., Chakrabarti, A., Kumari, S., Bhatia, A., Banerjee, D., 2016. Antiapoptotic and neuroprotective role of curcumin in Pentylenetetrazole (PTZ) induced kindling model in rat. Indian J. Exp. Biol. 54 (2), 133–141.

Saim, N.a., Dean, J.R., Abdullah, M.P., Zakaria, Z., 1997. Extraction of polycyclic aromatic hydrocarbons from contaminated soil using Soxhlet extraction, pressurised and atmospheric microwave-assisted extraction, supercritical fluid extraction and accelerated solvent extraction. J. Chromatogr. A 791 (1), 361–366.

Saklani, N., Purohit, V.K., Andola, H.C., Rana, M., Chauhan, R., Nautiyal, A., 2012. Biological activities of *Valeriana wallichii* D. Med. Plants: Int. J. Phytomed. Relat. Ind. 4 (3), 115–120.

Sasidharan, S., Chen, Y., Saravanan, D., Sundram, K.M., Latha, L.Y., 2011. Extraction, isolation and characterization of bioactive compounds from plants' extracts. African J. Trad. Complem. Alt. Med. 8 (1), 1–10.

Saxena, V., Chourasia, S., 2001. A new isoflavone from the roots of *Asparagus racemosus*. Fitoterapia 72 (3), 307–309.

Scalbert, A., Duval, L., Peng, S., Monties, B., Du Penhoat, C., 1990. Polyphenols of Quercus robus L.: II. Preparative isolation by low-pressure and high-pressure liquid chromatography of heartwood ellagitannins. J. Chromatogr. A 502 (1990), 107–119.

Seger, C., Sturm, S., Stuppner, H., 2013. Mass spectrometry and NMR spectroscopy: modern high-end detectors for high resolution separation techniques: state of the art in natural product HPLC-MS, HPLC-NMR, and CE-MS hyphenations. Nat. Prod. Rep. 30 (7), 970–987.

Sharma, P., Kirar, V., Meena, D.K., Suryakumar, G., Misra, K., 2012. Adaptogenic activity of Valeriana wallichii using cold, hypoxia, and restraint multiple stress animal model. Biomed. Aging Pathol. 2 (4), 198–205.

Siddiqui, M.R., AlOthman, Z.A., Rahman, N., 2017. Analytical techniques in pharmaceutical analysis: a review. Arab. J. Chem. 10 (s1), 1409–S1421.

Singh, B., Chaurasia, O.P., 1998. Medicinal flora of Indian cold desert. In: Proc. XXV International Horticultural Congress, Part 13: New and Specialized Crops and Products, Botanic Gardens and Human-Horticulture Relationship. Vol. 523. pp. 65–74.

Singh, B., Sidiq, T., Joshi, P., Jain, S.K., Lawaniya, Y., Kichlu, S., Khajuria, A., Vishwakarma, R.A., Bharate, S.B., 2013. Anti-inflammatory and immunomodulatory flavones from Actinocarya tibetica Benth. Nat. Prod. Res. 27 (23), 2227–2230.

Srimal, R., 2001. Database of medicinal plants. Curr. Sci. 30 (3), 463.

Sultan, P., Jan, A., Pervaiz, Q., 2016. Phytochemical studies for quantitative estimation of iridoid glycosides in Picrorhiza kurroa Royle. Bot. Stud. 57 (1), 1–7.

Theis, N., Lerdau, M., 2003. The evolution of function in plant secondary metabolites. Int. J. Plant Sci. 164 (S3), 93–102.

Tripathi, A., Prajapati, V., Verma, N., Bahl, J., Bansal, R., Khanuja, S.P.S., Kumar, S., 2002. Bioactivities of the leaf essential oil of *Curcuma longa* (var. ch-66) on three species of stored-product beetles (Coleoptera). J. Econ. Entomol. 95 (1), 183–189.

Verma, P., Gauttam, V., Kalia, A.N., 2014. Comparative pharmacognosy of Pashanbhed. J. Ayurveda Integr. Med. 5 (2), 104.

Villanueva-Bermejo, D., Zahran, F., Garcia-Risco, M. R., Reglero, G. and Fornari, T., 2017. Supercritical fluid extraction of Bulgarian Achillea millefolium. J. Supercrit. Fluids, 119, pp.283–288. https://doi.org/10.1016/j.supflu.2016.10.005.

Wang, J.-X., Xiao, X.-H., Li, G.-K., 2008. Study of vacuum microwave-assisted extraction of polyphenolic compounds and pigment from Chinese herbs. J. Chromatogr. A 1198–1199 (2008), 45–53.

Wang, W., 2000. Lyophilization and development of solid protein pharmaceuticals. Int. J. Pharm. 203 (1), 1–60.

Wangchuk, P., 2014. Phytochemical Analysis, Bioassays, and the Identification of Drug Lead Compounds From Seven Bhutanese Medicinal Plants. PhD thesis, University of Melbourne.

Wijnsma, R., Go, J., Van Weerden, I., Harkes, P.V., Verpoorte, R. and Svendsen, A.B., 1985. Anthraquinones as phytoalexins in cell and tissue cultures of Cinchona species. Plant Cell Rep., 4(5), pp. 241–244.

Wu, L., Hu, M., Li, Z., Song, Y., Yu, C., Zhang, Y., Zhang, H., Yu, A., Ma, Q., Wang, Z., 2015. Determination of triazine herbicides in fresh vegetables by dynamic microwave-assisted extraction coupled with homogeneous ionic liquid microextraction high performance liquid chromatography. Anal. Bioanal. Chem. 407 (6), 1753–1762.

Xiong, X., Yin, Y., Huang, Y., Wang, Y., Wen, Q., Mu, Y., Shu, X., Zhan, Z., Zhou, Y. and Qiu, G., 2013. Methods of Amino Acid Analysis, Nutritional and Physiological Functions of Amino Acids in Pigs. Springer, Vienna, pp.217–229. https://doi.org/10.1007/978-3-7091-1328-8_15.

Yan, X., Wang, W., Zhang, L., Zhao, Y., Xing, D., Du, L., 2007. A high-performance liquid chromatography with UV-vis detection method for the determination of Brazilein in plant extract. J. Chromatogr. Sci. 45 (4), 212–215.

Yu, Y., Chen, B., Chen, Y., Xie, M., Duan, H., Li, Y., Duan, G., 2009. Nitrogen-protected microwave-assisted extraction of ascorbic acid from fruit and vegetables. J. Separation Sci. 32 (23–24), 4227–4233.

Further Reading

De Jong, G. (Ed.), 2016. Capillary Electrophoresis-Mass Spectrometry (CE–MS): Principles and Applications. John Wiley & Sons, pp. 1–368. ISBN: 978-3-527-33924-2.

Erickson, M.D., 1979. Gas chromatography/Fourier transform infrared spectroscopy applications. Appl. Spectrosc. Rev. 15 (2), 261–298.

Pal, M., Bhardwaj, A., Manickam, M., Tulsawani, R., Srivastava, M., Sugadev, R., Misra, K., 2015. Protective efficacy of the caterpillar mushroom, Ophiocordyceps sinensis (ascomycetes), from India in neuronal hippocampal cells against hypoxia. Int. J. Med. Mushrooms 17 (9), 829–840.